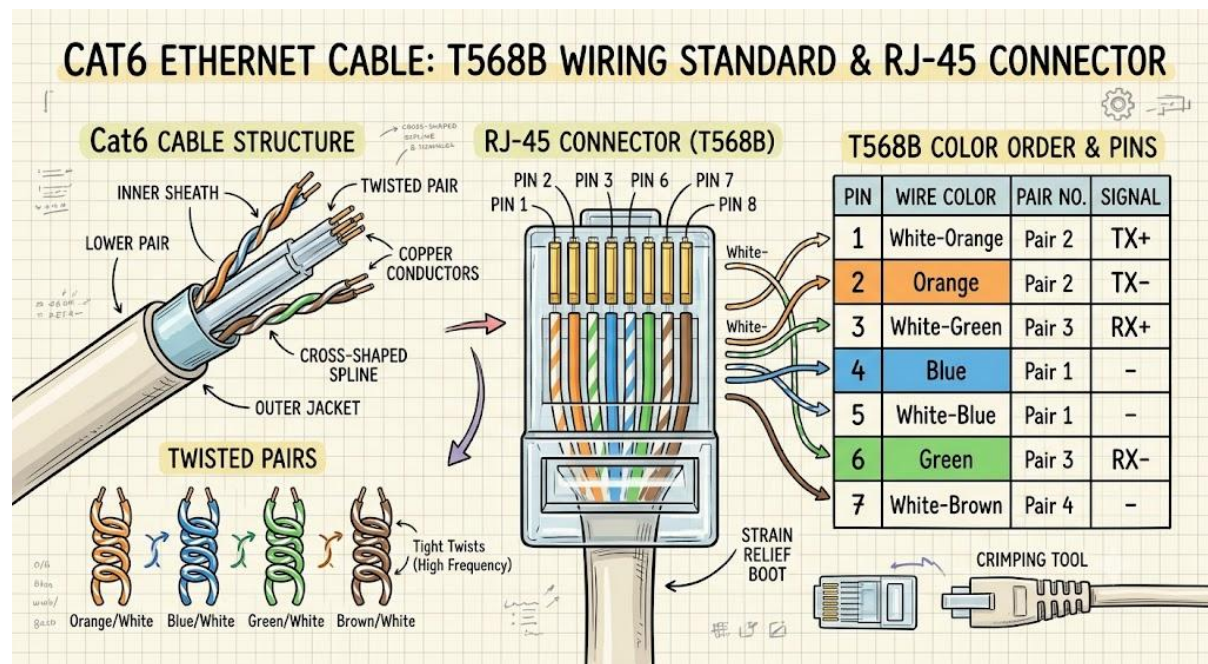


TASK-1



TASK-2

Network Cable	Real-World App/Device	Why It Is Used
Cat5e	WiFi Router	Suitable for home and small office networks, providing speeds up to 1 Gbps .
Cat6	Zomato Terminal	POS Supports faster speeds (up to 10 Gbps over short distances) for quick and reliable order processing.
Fiber Optic	IPL Server	Streaming Offers very high speed , long-distance communication, and low latency for smooth HD/live streaming.

TASK-3

Steps to Crimp an RJ-45 Connector**

1. Cut the Ethernet cable to the required length.
2. Strip about ****2–3 cm**** of the outer cable jacket using a cable stripper.
3. Untwist the wire pairs and arrange them in the ****T568B**** color order:

- * White-Orange
- * Orange
- * White-Green
- * Blue
- * White-Blue
- * Green
- * White-Brown
- * Brown

4. Straighten the wires and trim them evenly.
5. Insert the wires fully into the RJ-45 connector, ensuring each wire reaches the end.
6. Place the connector into the crimping tool and squeeze firmly.
7. Test the cable using a network cable tester to verify all connections are correct.

Two Common Mistakes to Avoid

1. ****Wrong wire order**** – Incorrect color sequence can prevent the cable from working.
2. ****Wires not fully inserted**** – If the wires do not reach the end of the RJ-45 connector, the connection may fail.

TASK-4

Gaming Zone Network Setup for an IPL Screening Event

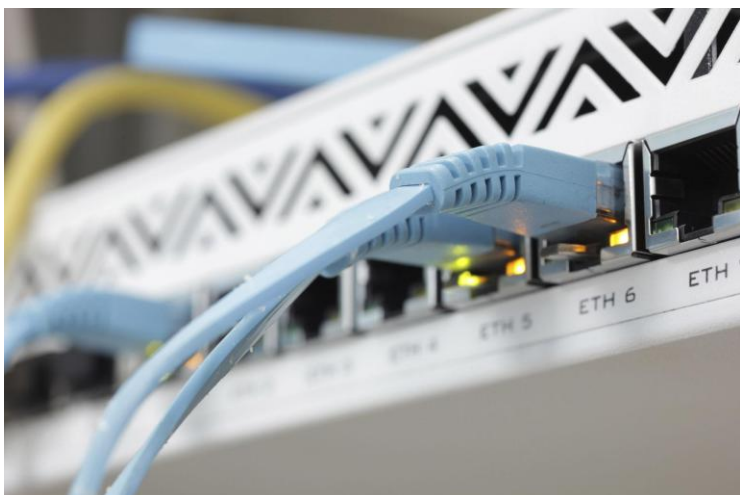
Connection	Cable Type	Connector Type	Reason
Internet to Switch	Cat6 Ethernet Cable	RJ-45	Provides high-speed (up to 10 Gbps) and reliable network connectivity.
Switch to Gaming Consoles	Cat6 Ethernet Cable	RJ-45	Ensures low latency and stable connections for multiplayer gaming.

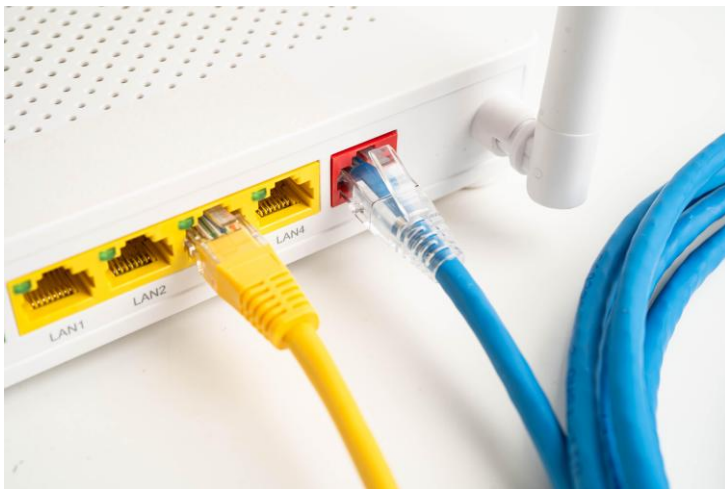
Connection	Cable Type	Connector Type	Reason
Switch to Smart TVs/Streaming PCs	Cat6 Ethernet Cable	RJ-45	Delivers smooth IPL streaming without buffering.
Long-distance Backbone (if needed)	Fiber Cable	Optic LC Connector	Supports very high speed over long distances with minimal signal loss.

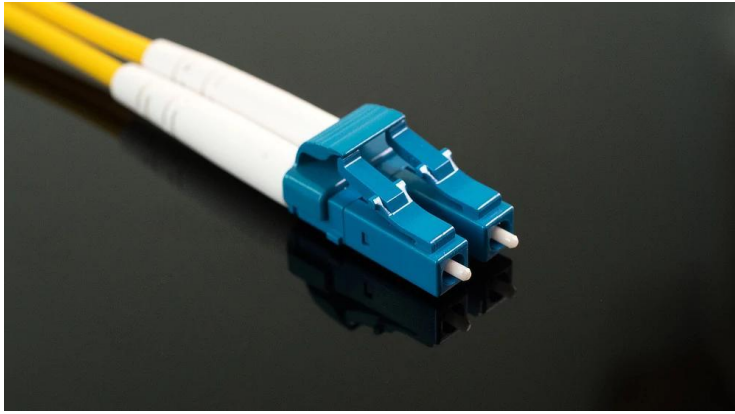
Explanation

I would use **Cat6 cables with RJ-45 connectors** to connect the gaming consoles, PCs, and smart TVs to the network switch because they provide fast speeds, low latency, and reliable performance. If the network needs to cover a long distance within the venue, I would use **fiber optic cables with LC connectors** for the backbone connection to ensure maximum speed and minimal signal loss.

TASK-5







Example Observation

Photo 1: RJ-45 Ethernet Cable (Cat6)

- **Type:** Cat6 Ethernet Cable
- **Connector:** RJ-45
- **Connected Devices:** WiFi Router ↔ Laptop/PC or Network Switch
- **Use:** Provides a fast and stable wired internet connection for homes, offices, and gaming setups.

Photo 2: RJ-45 Cable Connected to a Network Switch

- **Type:** Cat6 Ethernet Cable
- **Connector:** RJ-45
- **Connected Devices:** Network Switch ↔ Multiple Computers
- **Use:** Connects several computers on the same Local Area Network (LAN).

Photo 3: Fiber Optic Cable

- **Type:** Fiber Optic Cable
- **Connector:** LC Connector
- **Connected Devices:** ISP Modem/OLT ↔ Data Center or High-Speed Network Equipment
- **Use:** Used for high-speed internet over long distances with very low signal loss.